

Python in Electrical Engineering

PandaPower – Contingency analysis

1. Create the power system model in **PandaPower** using the exact specifications (bus data, generation, loads, and line parameters) from the previous class assignment.
2. Perform an **N-1 contingency analysis** by simulating the outage of each line individually.
3. For **each contingency** (line outage), do the following:
 - Temporarily set the line out of service (`in_service = False`)
 - Run the AC power flow calculation (`pp.runpp(net)`)
 - Record the following results:
 - Minimum bus voltage magnitude in the system (p.u.)
 - Maximum line loading (%) across all remaining lines
 - Whether the power flow converged or not
 - Restore the line to service (`in_service = True`)
4. From the contingency analysis results, identify and report:
 - **The most critical contingency for thermal overload** (the line outage that causes the **highest maximum line loading**)
 - **The most critical contingency for voltage violation** (the line outage that causes the **lowest minimum bus voltage**)
5. For the **most critical contingencies** determine the **smallest possible power source** that eliminates all violations under that outage. You may use **one** of the following devices:
 - A **shunt capacitor** (reactive power injection in Mvar) — recommended for voltage problems
 - An **additional active power source** (generator in MW)
6. Requirements:
 - Find the **minimal size** (smallest Mvar or MW) of the device required to remove all violations.
 - Determine the **best connection point** (bus number).
 - Test at least **buses 3, 4, and 5** as possible locations and compare the results.
 - After placing the device, re-run the power flow under the contingency and verify that:
 - All bus voltages ≥ 0.90 p.u.
 - All line loadings $\leq 120\%$
 - The power flow converges successfully

Deliverables

Submit the following:

1. Python code
2. Summary table of all N-1 contingencies showing:
 - Outaged line
 - Minimum voltage (p.u.)
 - Maximum line loading (%)
 - Convergence status
3. Mitigation solution including:
 - Type of device (shunt capacitor or generator)
 - Size (MW or Mvar)
 - Optimal connection bus
 - Post-mitigation results (voltages and loadings)
4. Short discussion (½ to 1 page) covering:
 - Why the identified contingency is the most severe
 - Why you chose that particular bus for compensation
 - Limitations of this mitigation approach